Unequal Job Opportunities and the Measurement of Welfare Inequality

Hisham Haydar

LISER and University of Luxembourg

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The same job, for two opposite reasons

one person

chose it
from a wide set

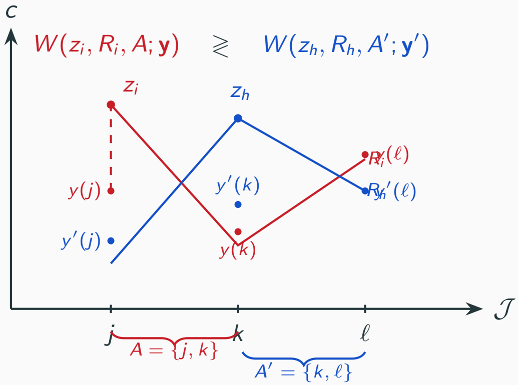
the same
observed job

the other

had almost
nothing else

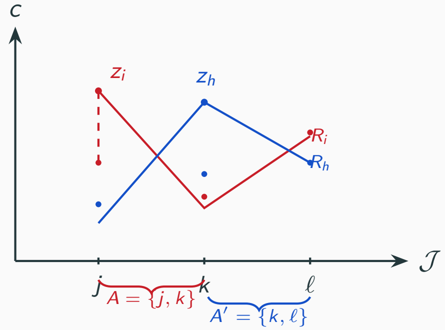
Whatever the choice set cannot say, the taste parameter says.

The theory puts jobs on the axis and splits the wage in two



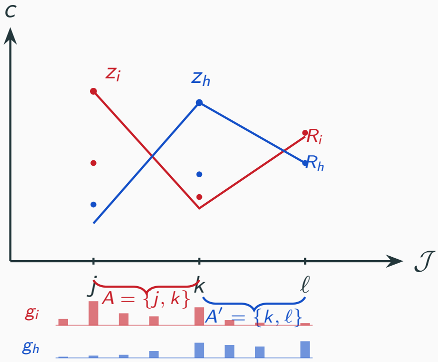
Haydar and Maniquet, the companion paper.

I estimate the set as a distribution, the pay as a density



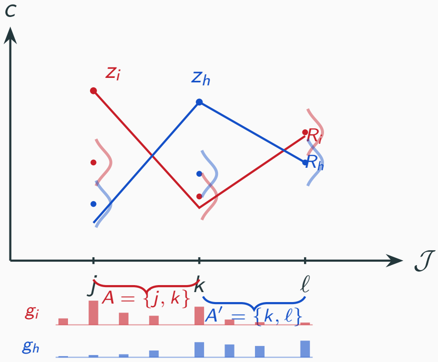
Two things change. Nothing else does.

I estimate the set as a distribution, the pay as a density



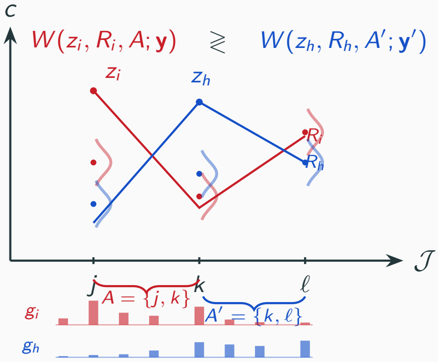
Estimated, not a list.

I estimate the set as a distribution, the pay as a density



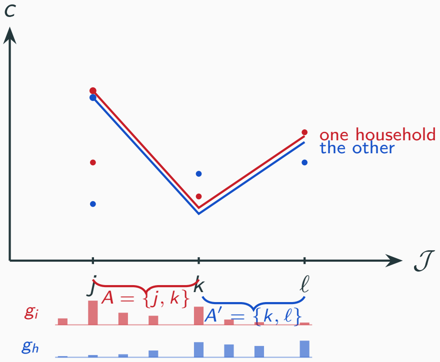
Every job carries a density, not a number.

I estimate the set as a distribution, the pay as a density



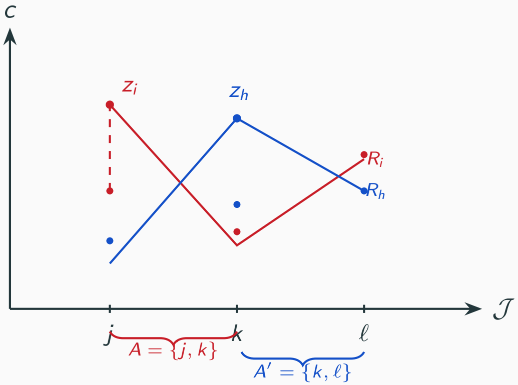
Every job carries a density, not a number.

Two real households sit here: same job, opportunity 0.851 against 0.349



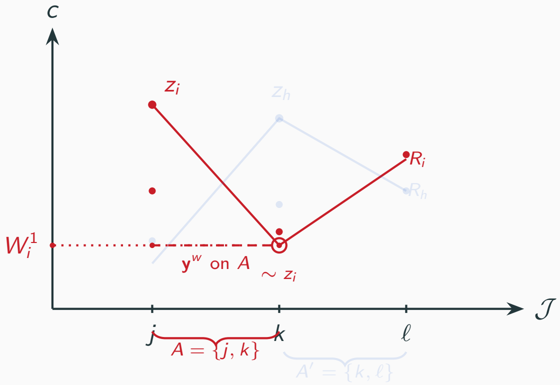
Same segments. Different distributions.

Equal pay on the set you can reach, then read off the level



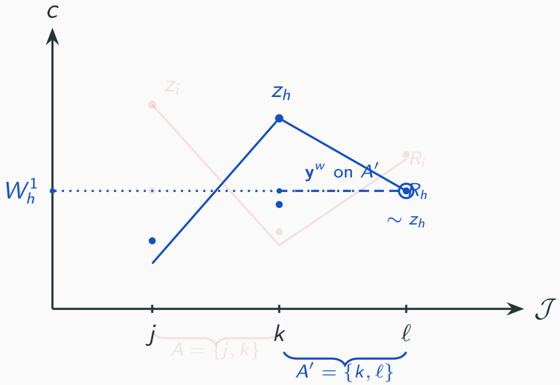
The measure the paper takes to the data.

Equal pay on the set you can reach, then read off the level



Equal pay on the feasible set.

Equal pay on the set you can reach, then read off the level



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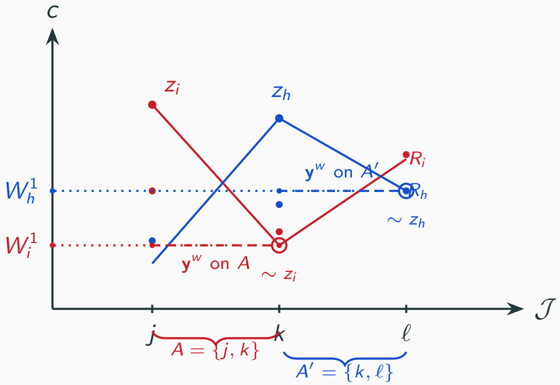
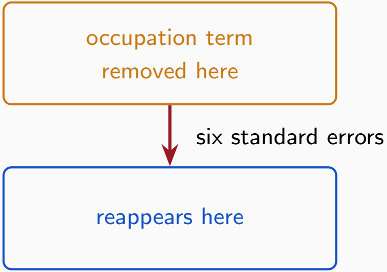


Empirically: a common reference pay.

Drop the opportunity object and the effect moves, it does not go



The variation goes wherever the model leaves a door open.

Availability is four factors, estimated with preferences at once

$$g = \underbrace{g^{E}}_{\text{whether work is available}} \; \underbrace{g^{H}}_{\text{at what hours}} \; \underbrace{g^{\mathrm{Occ}}}_{\text{in which occupation}} \; \underbrace{g^{W}}_{\text{at what pay}}$$

The hours factor is the same for every household. I say so.

Each of 1,555 households gets its observed job plus 100 drawn ones

$$V_{ij} = u_{ij} + \log g_{ij} - \log q_{ij}$$

preferences opportunity sampling correction

q is computation. g is economics.

Every job is priced.

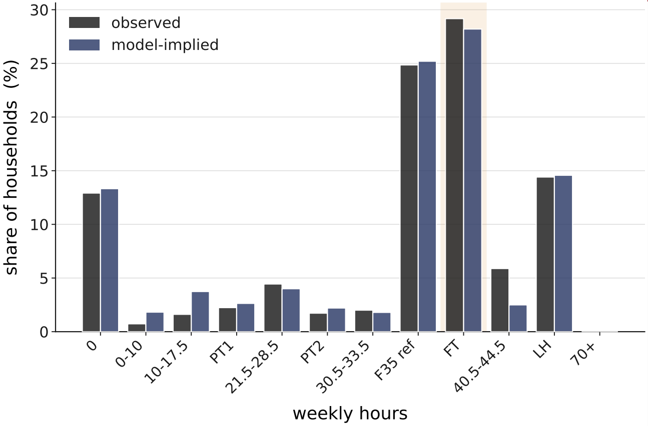
Two things I cannot separate, and I say so

access *versus* capability

persistent household heterogeneity

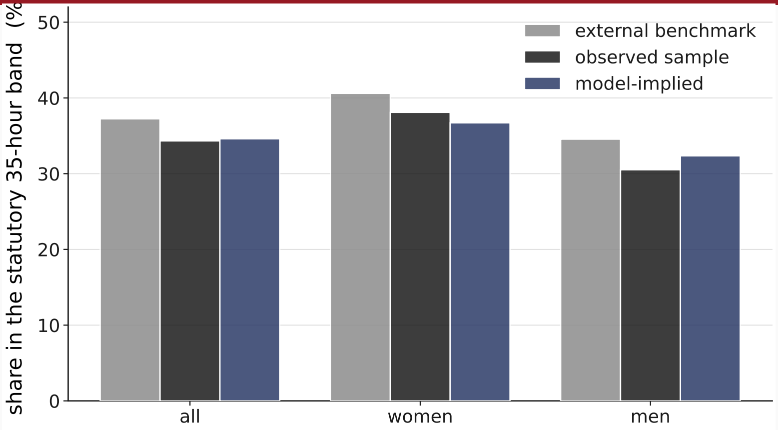
This is a decomposition. It is not a causal estimate.

The model reproduces the statutory week: 25.20% against 24.85% observed



A peak in what is on offer. Not the statute's effect.

A source the model never saw puts the band at 37.2%



Validation only. No outside number enters the likelihood.

Two orders, averaged: that is the whole decomposition

	own preferences	reference preferences
own opportunities	$I_{00} = 0.1343$	$I_{10} = 0.1483$
reference opportunities	$I_{01} = 0.0310$	$I_{11} = 0.0000$

Equalising preferences alone raises inequality.

Two orders, averaged: that is the whole decomposition

	own preferences	reference preferences
own opportunities	$I_{00} = 0.1343$	$I_{10} = 0.1483$
reference opportunities	$I_{01} = 0.0310$	$I_{11} = 0.0000$

$$C_{\mathrm{pref}} = \tfrac{1}{2}[(I_{00} - I_{10}) + (I_{01} - I_{11})] = 0.0085$$

$$C_{\mathrm{env}} = \tfrac{1}{2}[(I_{00} - I_{01}) + (I_{10} - I_{11})] = 0.1258$$

The Shapley value averages both orders.

Two orders, averaged: that is the whole decomposition

job access	0.020
earning opportunities	0.028
household endowments and needs	0.078

Inside opportunities and budgets, the Owen value.

Two orders, averaged: that is the whole decomposition

job access	0.020
earning opportunities	0.028
household endowments and needs	0.078
of which geographic	0.018

Almost all of job access is geographic.

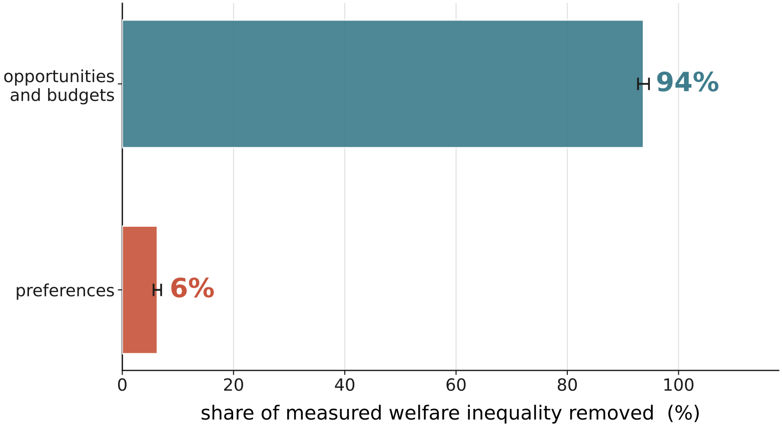
The non-preference environment carries 94% of welfare inequality

$$\underbrace{C_{\mathrm{pref}}}_{\text{preferences}} + \underbrace{C_{\mathrm{env}}}_{\text{opportunities and budgets}} = I_{00} - I_{11}$$

$$C_{\mathrm{env}} = \underbrace{C_{\mathrm{acc}}}_{\text{job access}} + \underbrace{C_{\mathrm{earn}}}_{\text{earning opportunities}} + \underbrace{C_{\mathrm{needs}}}_{\text{endowments and needs}}$$

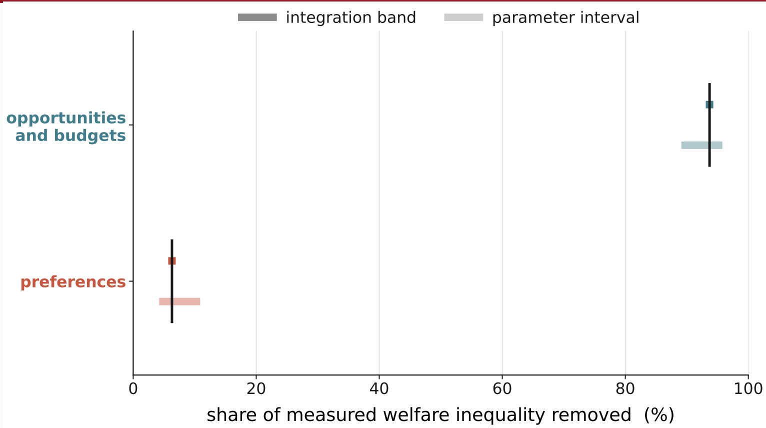
The fully common state is numerically zero — a tested property.

The non-preference environment carries 94% of welfare inequality



Bars are integration precision, not confidence intervals.

The non-preference environment carries 94% of welfare inequality



Two kinds of interval, on offset rows. Never merged.

How much of welfare inequality is unequal job opportunities?

job access and earning opportunities	$35.51 \pm 0.95\%$
the whole non-preference environment	$93.70 \pm 0.61\%$
preferences	6.30%

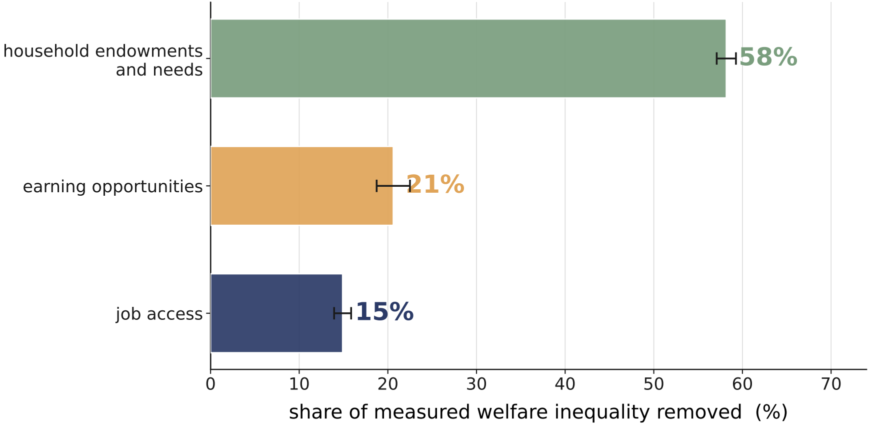
Opportunities and budgets carry almost all of it.

How much of welfare inequality is unequal job opportunities?

preferences share	$6.3\% \to 6.4\%$
measured inequality	-24%
relabelled to household endowments and needs	36%
out of the total	68%
the sex gap in tastes	$+0.43 \to -1.99$

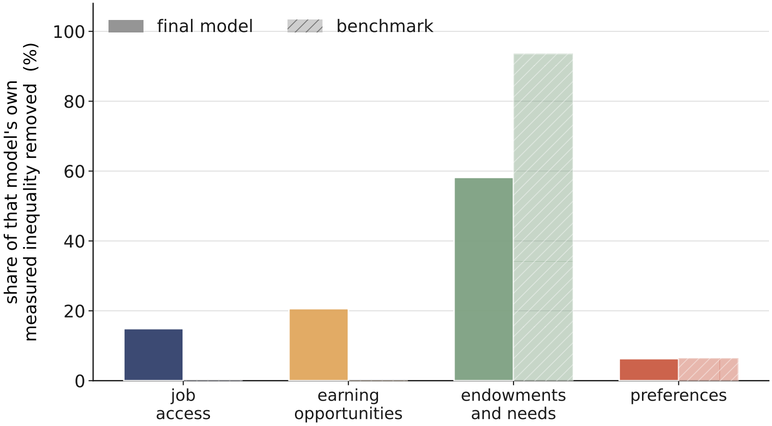
Omitted opportunities are relabelled, not turned into taste.

Budgets carry 58%, job opportunities 36%



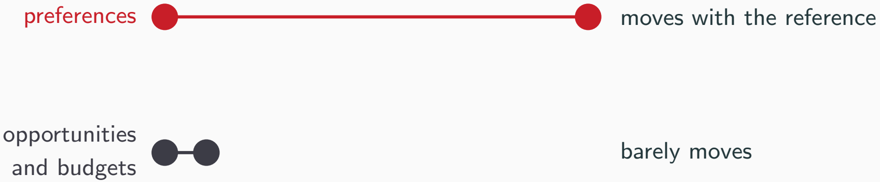
I claim no ordering between job access and earnings.

The same result, drawn: the zeros are the benchmark's definition



Job access and earnings are exactly zero here. By construction.

The preference share is the fragile quantity: 6% to 11%



No sign changes. No ordering changes. Anywhere.

Couples give the same order and a preference share I do not trust

Singles stay the headline. Couples are the companion.

Separate what people want from what they can get

Method	every job priced, opportunities estimated
Finding	opportunities and budgets carry 94%
Margin	preferences 6% to 11%

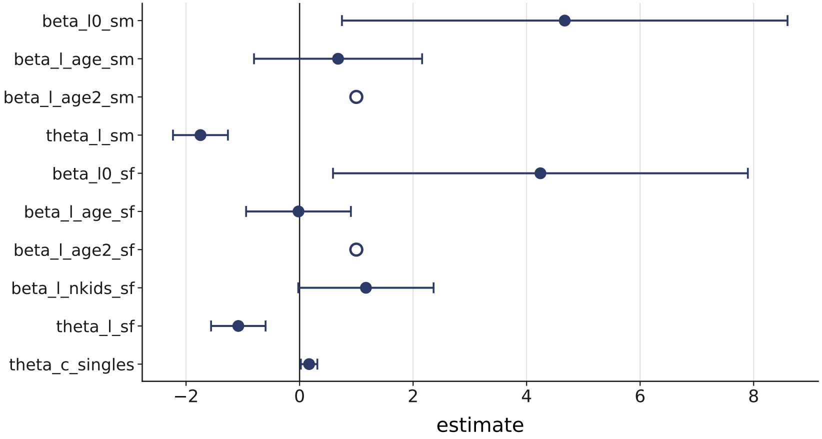
B0 — The preference block

$$u_{ij} \;=\; \beta_\ell(x_i)\,\mathrm{BC}(\ell_{ij};\theta_\ell) \;+\; \mathrm{BC}(c_{ij};\theta_c)$$

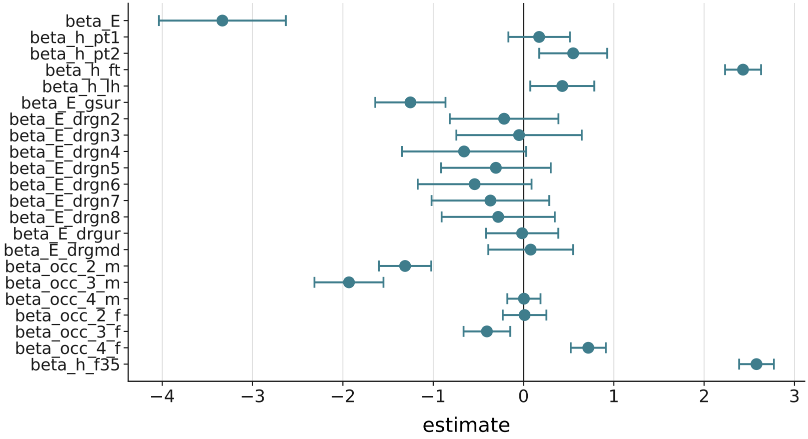
leisure · consumption

Box–Cox in both. Consumption fixes the scale.

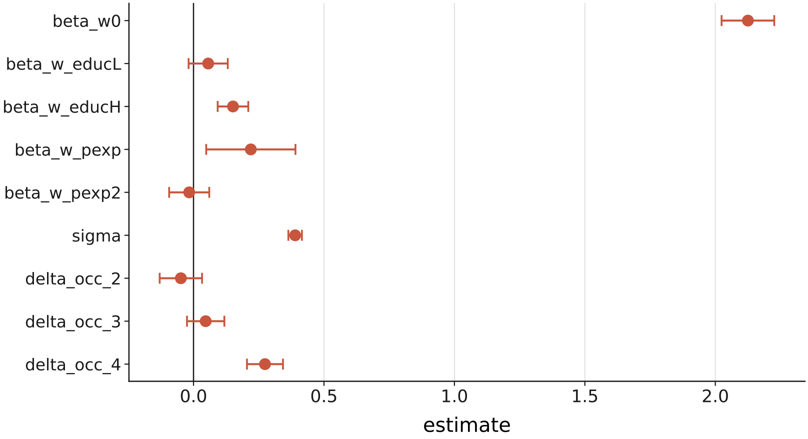
B1 — The estimated coefficients: preferences



B1 — The estimated coefficients: job access



B1 — The estimated coefficients: earning opportunities



B2 — Exhaustiveness is a tested property

defect	what fixed it
a missing budget channel	a fourth player from its own marginals
asymmetric money-metric inversion	the reference side moves with the coalition
an incomplete male reference block	a structural zero on the child term

The fully common state is 0.000000 — numerically zero.

B3 — The scale moves with the channel that equalises needs

	raw	equivalized
preferences	6.30%	7.98%
opportunities and budgets	93.70%	92.02%
job access	14.90%	9.68%
earning opportunities	20.62%	16.24%
household endowments and needs	58.18%	66.10%
market side	35.51%	25.92%

Freezing the own scale leaves exactly the Gini of its inverse.

B4 — I tried the hours-wish moment; there is no counterpart

ground	**what it shows**
definitional	the candidate's optimum tracks pay
identification	the hours block moves the moment by **exactly 0**
numerical	the level can agree; the profile cannot

Tried, and it fails on three independent grounds.

B5 — Where the welfare family comes from

Compensation

unequal productivity should not become unequal well-being

Responsibility

preference-based choices should not be compensated

$$\nexists\, W: \quad \text{Full Compensation} \;\wedge\; \text{Full Responsibility}$$

[Fleurbaey & Maniquet, *SCW* 2005]

W^1 is one compromise — the one this paper takes to the data.

Haydar and Maniquet, *Jobs and Well-Being Measurement* (companion paper).